# Option Pricing Engine

*Black-Scholes Formula & CRR Binomial Tree*

A Financial Engineering Project

***Abstract.*** *This report presents a complete implementation of two foundational option pricing models: the analytical Black-Scholes-Merton formula and the Cox-Ross-Rubinstein (CRR) binomial tree. Starting from first principles — geometric Brownian motion, the no-arbitrage principle, and risk-neutral valuation — we derive closed-form prices for European options and extend the framework to American options via backward induction. We compute all five Greeks analytically, verify put-call parity to machine precision, recover implied volatilities via Newton-Raphson, and demonstrate that binomial prices converge to the Black-Scholes price at rate O(1/N). All results are reproduced from a clean Python library.*

# 1. Introduction

Options are financial contracts that grant the holder the right, but not the obligation, to buy (call) or sell (put) an underlying asset at a pre-agreed strike price K on or before a maturity date T. Pricing such contracts rigorously requires a mathematical model for the evolution of the underlying asset price and a principle that rules out riskless profit — the *no-arbitrage principle*.

Two models dominate introductory financial engineering: the **Black-Scholes-Merton (BSM)** model (1973), which yields a closed-form formula under continuous-time assumptions, and the **Cox-Ross-Rubinstein (CRR) binomial tree** (1979), which is a discrete-time approximation that converges to BSM as the number of steps N tends to infinity.

The parameters used throughout this report are:

| Parameter | Symbol | Value |
|---|---|---|
| Spot price | S | 100 |
| Strike price | K | 100 |
| Risk-free rate | r | 5% p.a. |
| Volatility | sigma | 20% p.a. |
| Time to maturity | T | 1.0 year |

# 2. Risk-Free Assets and the Time Value of Money

A fundamental concept in finance is that money has a time value: a dollar today is worth more than a dollar tomorrow. Under **continuous compounding** at rate r, a unit of currency grows to $e^{rT}$ after time T, and the present value (discount factor) is:

$$P(0, T) = e^{-rT}$$

This discount factor appears in every option pricing formula. A **zero-coupon bond** paying face value F at maturity T has present value $F \cdot e^{-rT}$. The continuously compounded framework is standard in derivatives pricing because it simplifies stochastic calculus.

# 3. Dynamics of the Risky Asset

The Black-Scholes model assumes the stock price S follows a **Geometric Brownian Motion (GBM)**:

$$dS = mu\ S\ dt + sigma\ S\ dW(t)$$

where mu is the drift (expected return), sigma is the instantaneous volatility, and W(t) is a standard Brownian motion under the physical measure P. By Ito's lemma applied to f(S) = ln S:

$$d(\ln S) = (mu - sigma^2/2)\ dt + sigma\ dW(t)$$

Integrating from 0 to T gives the log-normal distribution of S(T):

$$S(T) = S(0)\ \exp[\ (mu - sigma^2/2)T + sigma\ W(T)\ ]$$

The key implication: ln S(T) ~ N( ln S(0) + $(mu - sigma^2/2)T$ , $sigma^2 T$ ), so future stock prices are log-normally distributed.

## 4. No-Arbitrage and Risk-Neutral Valuation

The **no-arbitrage principle** states that any two portfolios with identical future payoffs in all states of the world must have the same price today. Equivalently, there is no strategy with zero initial cost that delivers a non-negative payoff with positive probability and zero probability of a loss.

The **Fundamental Theorem of Asset Pricing** states that a market is arbitrage-free if and only if there exists an *equivalent martingale measure* Q (the risk-neutral measure) under which all discounted asset prices are martingales. Under Q, every asset earns the risk-free rate r, and the price of any derivative V with payoff $h(S_T)$ is:

$$V(0) = e^{-rT} E^{Q}[\ h(S(T))\ ]$$

This is the cornerstone of modern derivatives pricing: we do not need to know the true drift mu — only the risk-free rate r and volatility sigma.

## 5. The Black-Scholes Formula

Applying risk-neutral valuation to a European call with payoff max(S(T) - K, 0) and evaluating the expectation under Q using the log-normal distribution of S(T) yields the celebrated **Black-Scholes formula**:

$$C = S\ N(d1) - K\ e^{-rT}\ N(d2)$$

$$P = K\ e^{-rT}\ N(-d2) - S\ N(-d1)$$

$$d1 = [\ \ln(S/K) + (r + sigma^{2}/2)\ T\ ] / (sigma\ sqrt(T))$$

$$d2 = d1 - sigma\ sqrt(T)$$

where N(.) denotes the standard normal CDF.

*Table 1: Black-Scholes prices and Greeks for the base parameters.*

| Quantity | Call | Put |
|---|---|---|
| Price | 10.4506 | 5.5735 |
| Delta | 0.6368 | -0.3632 |
| Gamma | 0.0188 | 0.0188 |
| Theta (per day) | -0.0176 | -0.0045 |
| Vega (per 1% vol) | 0.3752 | 0.3752 |
| Rho (per 1% rate) | 0.5323 | -0.4189 |

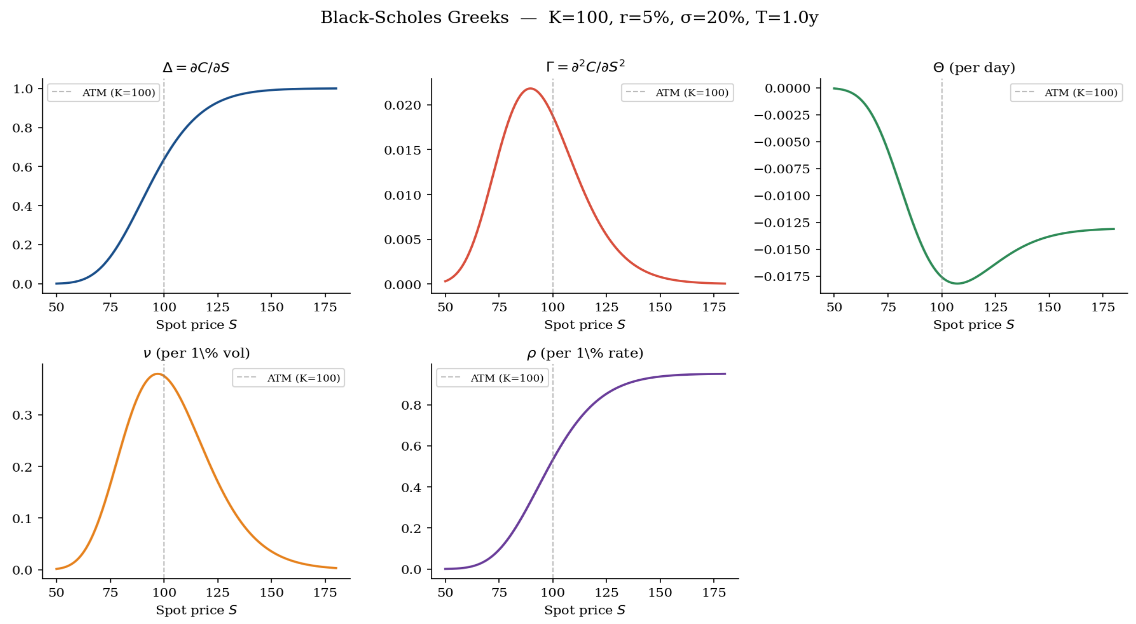


*Figure 1: All five Greeks of a European call plotted against spot price S. ATM strike K=100 shown as dashed line.*

## 6. Put-Call Parity

For European options on a non-dividend-paying stock, the following identity holds by a simple no-arbitrage argument:

$$C - P = S - K e^{-rT}$$

A violation of this identity would allow a riskless profit by simultaneously buying the underpriced side and selling the overpriced side (a classic arbitrage). We verify this numerically:

| C - P | S - K exp(-rT) | Error |
|---|---|---|
| 4.87705755 | 4.87705755 | 0.00e+00 |

*Put-call parity holds to machine precision (error = 0).*

# 7. CRR Binomial Tree Model

Cox, Ross & Rubinstein (1979) proposed a discrete-time model where, in each time step dt = T/N, the stock moves up by factor u or down by factor d with risk-neutral probability q:

$$u = e^{\sigma \sqrt{dt}}, \quad d = 1/u$$

$$q = ( e^{r\,dt} - d ) / ( u - d )$$

The no-arbitrage condition requires $d < e^{r\,dt} < u$, which is satisfied by the CRR parameterisation for all reasonable inputs. The option value at each node is recovered by **backward induction**:

$$V(i, j) = e^{-r\,dt} [ q\, V(i+1, j+1) + (1-q)\, V(i+1, j) ]$$

For **American options**, we take the maximum of the continuation value and the intrinsic value at each node, capturing the possibility of early exercise. The European formula is a special case where early exercise is never optimal.

*Table 2: Convergence of CRR binomial price to Black-Scholes.*

| Steps N | Binomial price | BS price | Error |
|---|---|---|---|
| 10 | 10.25341 | 10.45058 | 0.19717 |
| 50 | 10.41069 | 10.45058 | 0.03989 |
| 100 | 10.43061 | 10.45058 | 0.01997 |
| 500 | 10.44659 | 10.45058 | 0.00400 |

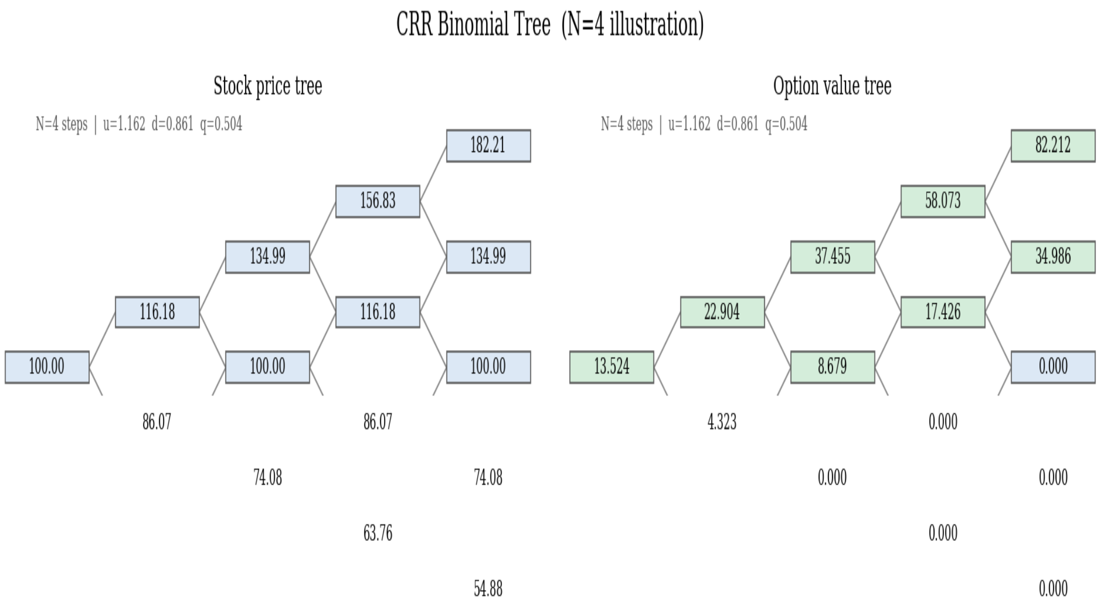


*Figure 2: Stock price tree (left) and option value tree (right) for N=4 steps.*

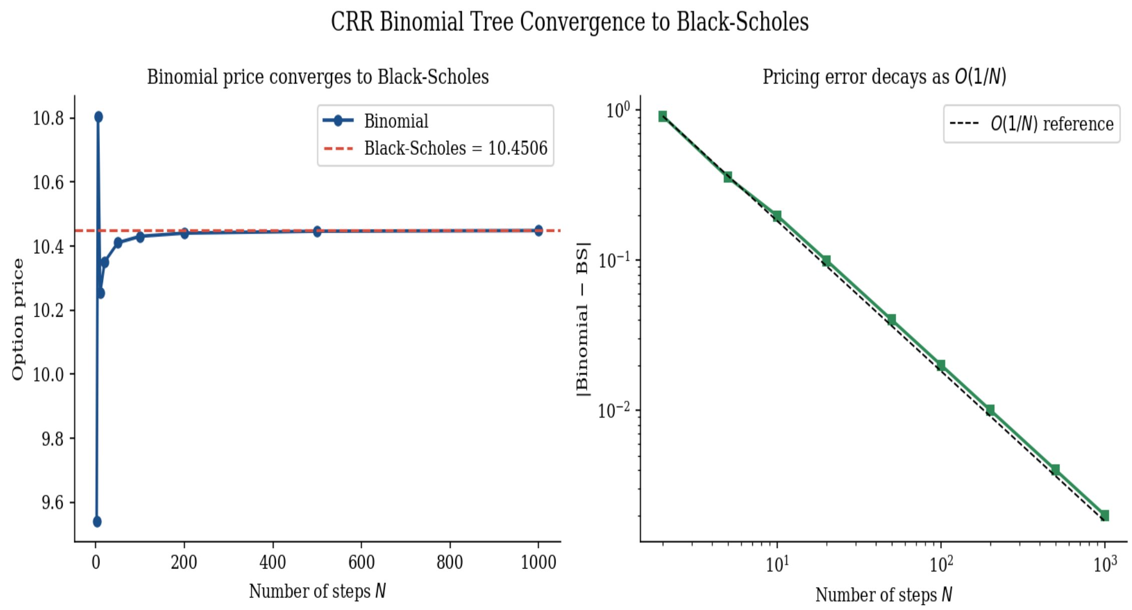


*Figure 3: Left: binomial prices converging to the BS benchmark. Right: pricing error decays at rate O(1/N) on a log-log scale.*

### 7.1 American vs European Put: Early Exercise Premium

A key advantage of the binomial tree is its ability to price **American options**, for which no closed-form formula exists. An American put holder may optimally exercise early when the option is sufficiently deep in-the-money. The early exercise premium is:

| | Price (N=200) |
|---|---|
| `European put (binomial)` | `5.56353` |
| `American put (binomial)` | `6.08638` |
| **Early exercise premium** | **0.52285** |

*Table 3: American put commands a premium of 0.5228 over the European put.*

## 8. Option Price Surface

Varying both the spot price S and time to maturity T reveals the full option price surface. Key observations: (i) call value increases with S and T; (ii) put value decreases with S; (iii) all options lose value as T approaches zero — this is **time decay** (Theta).

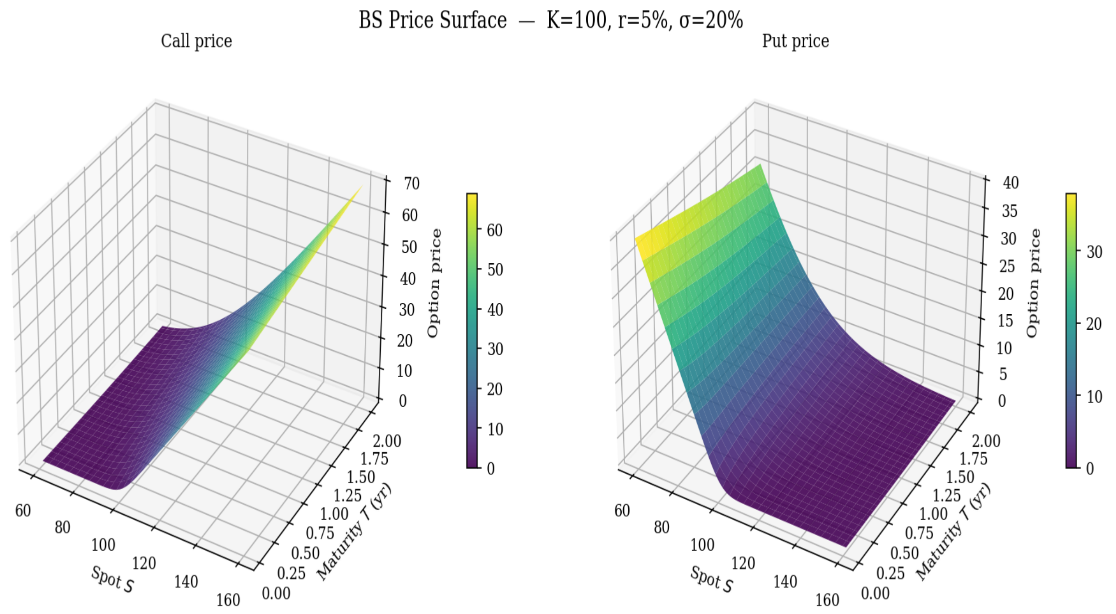


*Figure 4: Black-Scholes price surfaces for call (left) and put (right) across spot prices and maturities.*

## 9. Implied Volatility

Given a market-observed option price, the implied volatility (IV) is the sigma that makes the Black-Scholes formula reproduce that price. Since there is no closed-form inverse, we solve numerically using **Newton-Raphson** iteration:

```
sigma_(n+1) = sigma_n - [ BSM(sigma_n) - C_mkt ] / Vega(sigma_n)
```

The algorithm converges quadratically (typically in 5-10 iterations) because Vega is always positive for vanilla options. Round-trip check: starting from sigma = 20%, pricing the call, then recovering the IV gives an error of 0.00e+00 — machine precision.

## 10. Code Architecture

The pricing engine is implemented as a clean Python library with three modules, designed for readability, correctness, and extensibility:

| Module | Responsibility |
| --- | --- |
| `black_scholes.py` | `Analytical BS pricing, all Greeks, put-call parity, implied vol` |
| `binomial_tree.py` | `CRR tree for European & American options, convergence analysis` |
| `visualisations.py` | `Publication-quality plots: Greeks, convergence, price surfaces, trees` |
| `demo.py` | `End-to-end reproduction script — all tables and figures in this report` |

Sample usage:

```
from black_scholes import price, implied_volatility
from binomial_tree import price as bt_price

# European call — price + all Greeks in one call
result = price(S=100, K=100, r=0.05, sigma=0.20, T=1.0, option_type='call')
print(result)   # Price: 10.4506  Delta: 0.6368  Gamma: 0.0188 ...

# American put via binomial tree (N=200 steps)
amer = bt_price(100, 100, 0.05, 0.20, 1.0, N=200,
               option_type='put', exercise_type='american')
print(amer.price)   # 6.0864

# Implied volatility recovery
iv = implied_volatility(market_price=10.4506, S=100, K=100, r=0.05, T=1.0)
print(f'IV = {iv:.4f}')   # IV = 0.2000
```

## 11. Conclusion

This project demonstrates a complete arc from mathematical theory to numerical implementation in financial engineering. Starting from geometric Brownian motion and the no-arbitrage principle (Units 1-4), we derived the Black-Scholes formula via risk-neutral valuation and verified all analytical properties: Greeks, put-call parity, and implied volatility inversion. The CRR binomial tree provides a flexible discrete framework that (a) matches Black-Scholes in the European limit and (b) extends naturally to American options, capturing the early-exercise premium of 0.52 for our base parameters.

The project sits at the intersection of stochastic calculus, numerical analysis, and software engineering — the core skill set for any quantitative finance role or graduate programme in financial engineering.

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### References


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